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# DEPARTMENT OF MATHEMATICS

## MIDTERM TEST #1

## MTH 240 – Calculus II

Last Name (Print): [redacted] First Name: [redacted] Student Number: [redacted]

Signature: [redacted]

Date: Feb. 3, 2012, 4:15 pm

Duration: 1.5 hour

Version **B**

**Instructions:**

**Section (circle one)**

| | |
|---|---|
| Dr. C. Kim : | 1 2 3 4 5 |
| Dr. J. Pascal : | 6 7 8 9 10 |
| Dr. B. Tasic : | 12 13 14 15 |
| Dr. N. Ruzgar : | 16 17 18 19 20 |

**For Instructor's use only.**

1. This is a closed-book test. **Notes, calculators and other aids are not permitted.**
2. Verify that your test has pages 1-8.
3. (a) Unless otherwise instructed, **make sure you include all significant steps in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution.** Put a box around your final **answer.**

   (b) Write your solutions in the space provided. If you need more space, use the back of the page. Indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there. Marks (out of 50) are shown in brackets.
4. Do not separate the sheets.
5. Have your student card available on your desk.

| Page | Mark |
|---|---|
| 2 | 9 / |
| 3 | 7 / |
| 4 | 6 / |
| 5 | 5 / |
| 6 | 7 / |
| 7 | 7 / |
| 8 | 7 / |
| Total | 48 /50 |

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1. [3 marks] Write out the form of the partial fraction decomposition of the following function. **Do not evaluate the coefficients.**

$$f(x) = \frac{x}{(x+1)^2(x^2+1)(x^4+2x^2+1)} = \frac{x}{(x+1)^2(x^2+1)(x^2+1)(x^2+1)}$$

$$= \frac{x}{(x+1)^2(x^2+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{Cx+D}{x^2+1} + \frac{Fx+E}{(x^2+1)^2} + \frac{Gx+H}{(x^2+1)^3}$$

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2. [7 marks] Determine whether each of the following integrals is convergent or divergent. Evaluate those that are convergent.

$$\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx$$

$$\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx = \lim_{t\to\infty}\int_1^t \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx$$

$$= -2\lim_{t\to\infty}\left[e^{-u}\right]_1^t$$

$$= -2\lim_{t\to\infty}\left(e^{-t} - e^{-1}\right)$$

$$= -2\left[\lim_{t\to\infty} e^{-t}\right] + 2e$$

$$\left(e^{-\infty} = 0\right)$$

$$= -2.0 + 2e = 2e$$

$$\int \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx$$

$$= \text{let } u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}}dx$$

$$2\int e^{-u}.du$$

$$= -2e^{u} + c$$

$$x = \infty \to u = \infty$$

$$x = 1 \to u = 1$$

$$\frac{1}{2\sqrt{x}}$$

$$(x)^{\frac{1}{2}}$$

$$\boxed{\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}dx = 2e \quad \text{Conv.}}$$

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3. [7 marks] Determine whether each of the following integrals is convergent or divergent. Evaluate those that are convergent.

$$\int_0^3 \frac{2\,dx}{(x-1)^{\frac{2}{3}}}$$

undefined at 1

$$\int_0^3 \frac{2\,dx}{(x-1)^{2/3}} = \int_0^1 \frac{2\,dx}{(x-1)^{3/2}} + \int_1^3 \frac{2\,dx}{(x-1)^{2/3}}$$

$-\frac{2}{3}+1$

$\frac{-2+3}{3}$

$\frac{1}{3}$

$$= \lim_{t\to 1^-} \int_0^t \frac{2\,dx}{(x-1)^{3/2}} + \lim_{t\to 1^+} \int_t^3 \frac{2\,dx}{(x-1)^{2/3}}$$

$$= 6\left[\lim_{t\to 1^-} \left[(x-1)^{1/3}\right]_0^t + \lim_{t\to 1^+} \left[(x-1)^{1/3}\right]_t^3\right]$$

$$= 6\left[\lim_{t\to 1^-} (t-1)^{1/3} + 1 + \lim_{t\to 1^+} 2^{1/3} - \left[(t-1)^{1/3}\right]\right]$$

$$= 6\left[+1 + 2^{1/3}\right] = -6 + 6*2^{1/3} \quad \#$$

Conv.

$$\boxed{\int_0^3 \frac{2\,dx}{(x-1)^{2/3}} = +6 + 6*2^{1/3}}$$

$\int \frac{2\,dx}{(x-1)^{2/3}}$

let

$u = x-1$

$du = dx$

$2\int \frac{1}{u^{2/3}}\,du$

$2\int u^{-2/3}\,du$

$= \frac{2}{1/3} u^{1/3} + C$

$= 6(x-1)^{1/3} + C$

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4. [6 marks] Evaluate the following integral.

$$\int_{\sqrt{\frac{\pi}{2}}}^{\sqrt{\pi}} x^3 \cos(x^2)\, dx$$

$$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\pi} t \cdot \cos(t)\, dt$$

let
$t = x^2$
$dt = 2x\,dx$

$x \to \sqrt{\pi},\ t \to \pi$
$x \to \sqrt{\frac{\pi}{2}},\ t \to \frac{\pi}{2}$

by parts

$u = t$ $\quad dv = \cos(t)dt$
$du = dt$ $\quad v = \sin(t)$

$$\int t \cdot \cos(t)\, dt = t \cdot \sin(t) - \int \sin(t)\, dt$$

$$= t \cdot \sin(t) + \cos(t) + C$$

$$\frac{1}{2}\int_{\pi/2}^{\pi} \cos(t)\, t \cdot dt = \frac{1}{2}\Big[t \cdot \sin(t) + \cos(t)\Big]_{\frac{\pi}{2}}^{\pi}$$

$$= \frac{1}{2}[1] - \frac{1}{2}\left[\frac{\pi}{2} + 0\right]$$

$$= \boxed{\frac{1}{2} - \frac{\pi}{4}}$$

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$\sin\theta = \frac{1}{2}(1 - \cos 2\theta)$



5. [6 marks] **Evaluate the following integral.**

$$\int_0^{\frac{\pi}{2}} (2 - \sin\theta)^2 d\theta$$

-1

-4

$$\int_0^{\frac{\pi}{2}} (4 - 2\sin\theta + \sin^2\theta)\, d\theta$$

$$= \int_0^{\frac{\pi}{2}} 4\, d\theta - 2\int_0^{\frac{\pi}{2}} \sin\theta\, d\theta + \int_0^{\frac{\pi}{2}} \sin^2\theta\, d\theta$$

$$= 4\theta\Big]_0^{\frac{\pi}{2}} + 2\cos\theta\Big]_0^{\frac{\pi}{2}} + \frac{1}{2}\int_0^{\frac{\pi}{2}} (1 - \cos 2\theta)\, d\theta$$

$$= 2\pi - 2 + \left[\frac{1}{2}\theta\right]_0^{\frac{\pi}{2}} - \left[\frac{1}{4}\sin 2\theta\right]_0^{\frac{\pi}{2}}$$

$$= 2\pi - 2 + \frac{\pi}{4} + 0$$

$$= \boxed{2\left(\pi - 1 + \frac{\pi}{8}\right)}$$

#

$\frac{1}{4}\sin(\pi) - \frac{1}{4}\sin(0)$

$\frac{1}{4}$

$\theta$

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$\cos^2 + \sin^2 = 1$

$1 + \tan^2 = \sec^2$

$\tan^2 = \sec^2\theta - 1$

6. [7 marks] Evaluate the following integral.

$$\int \frac{x^2\,dx}{(x^2+2)^{\frac{3}{2}}}$$

(Hint: $\int \sec\theta\,d\theta = \ln|\sec\theta + \tan\theta| + C$)

Let $x = \sqrt{2}\tan\theta$,

$dx = \sqrt{2}\sec^2\theta\,d\theta$

$$\int \frac{2\tan^2\theta\,\sqrt{2}\sec^2\theta}{(\sqrt{2})^3 \cdot \sec^3\theta}\,d\theta$$

$$\int \frac{\tan^2\theta}{\sec\theta}\,d\theta$$

$$= \int \frac{(\sec^2\theta - 1)}{\sec\theta}\,d\theta = \int \sec\theta\,d\theta - \int \frac{1}{\sec\theta}\,d\theta$$

$$= \int \sec\theta\,d\theta - \int \cos\theta\,d\theta$$

$$\boxed{= \ln|\sec\theta + \tan\theta| - \sin\theta + C}$$

$\frac{7}{7}$

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$(u+1)(u+4)$

$u^2+u+4u+4$

7. [7 marks] Evaluate the following integral.

$$\int \frac{e^{2x}}{e^{2x}+5e^x+4}dx$$

let $u = e^x$

$du = e^x dx$

$$\int \frac{u}{u^2+5u+4}\,du$$

$$\int \frac{u}{(u+1)(u+4)}\,du$$

$$\frac{u}{(u+1)(u+4)} = \frac{A}{u+1} + \frac{B}{(u+4)}$$

$$u = A(u+4) + B(u+1)$$

let $u = -1, -4$

$-1 = 3A \Rightarrow \boxed{A = -\frac{1}{3}}$

$u = -4$

$-4 = -3B$

$\boxed{B = \frac{4}{3}}$

$$-\frac{1}{3}\int \frac{1}{u+1}\,du + \frac{4}{3}\int \frac{1}{u+4}\,du$$

$$= -\frac{1}{3}\left[\ln|u+1| - 4\ln|u+4|\right]$$

$$\boxed{\int \frac{e^{2x}}{e^{2x}+5e^x+4}\,dx = -\frac{1}{3}\left[\ln|e^x+1| - 4\ln|e^x+4|\right] + C}$$

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8. [7 marks] Evaluate the following integral.

$$\int \frac{x^{\frac{1}{2}}+2}{2(x^{\frac{3}{2}}+2x+5x^{\frac{1}{2}})}dx$$

let

$u = x^{1/2}$

$u^2 = x$

$2u\,du = dx$

$$\int \frac{(u+2)\cdot u\,du}{u^3+2u^2+5u}$$

$$\int \frac{(u+2)\cdot \cancel{u}}{\cancel{u}(u^2+2u+5)}du$$

$$\int \frac{u+2}{u^2+2u+5}du$$

$u^2+2u+1-1+5$

$(u+1)^2+4$

$$\int \frac{\frac{1}{2}(2u+2)+1}{u^2+2u+5}du = \frac{1}{2}\int \frac{2u+2}{u^2+2u+5}du + \int \frac{1}{u^2+2u+5}du$$

$$= \frac{1}{2}\ln|u^2+2u+5| + \int \frac{1}{(u+1)^2+4}du$$

$$= \frac{1}{2}\ln|u^2+2u+5| + \frac{1}{2}\tan^{-1}\left(\frac{u+1}{2}\right) + C$$

$$\boxed{= \frac{1}{2}\left[\ln|x+2\sqrt{x}+5| + \tan^{-1}\left(\frac{\sqrt{x}+1}{2}\right)\right] + C}$$

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